



Essay / Assignment Title: Predicting Car Prices Using Data Analytics and Machine Learning

Programme title: MSc Data Analytics

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CHAPTER ONE: Introduction

In this project, I applied data analytics and machine learning techniques to analyze an automobile dataset and predict car prices based on different vehicle features. The dataset contains several car attributes, including engine size, horsepower, fuel type, body style, normalized losses, engine type, city mileage, highway mileage, and price. The main target variable of this project is price.

The project started with data understanding, data cleaning, missing value handling, data type conversion, exploratory data analysis, feature engineering, binning, normalization, and visualization. After preparing the dataset, regression models such as Linear Regression and Random Forest Regressor were developed to predict car prices. The models were evaluated using R^2 score, Mean Absolute Error, and Mean Squared Error. In addition, a classification task was performed by converting car prices into Low, Medium, and High price categories and applying a Logistic Regression model.

The main objective of this project is to identify the key factors that influence car prices and to build predictive models that can support business decision-making and help customers estimate fair car prices. The code for this project is available at:

https://colab.research.google.com/drive/1rbtX1BQ9pTGKcYRU8WhSKds_6hbBnIs3?usp=sharing

CHAPTER TWO: Exploratory Data Analysis (EDA)

1.1. Dataset Loading and Column Header Assignment

The dataset used in this project is an automobile dataset (Schlimmer, 1985) containing information about different car attributes and their prices. The original dataset did not contain proper column headers, so the column names were manually assigned based on the dataset description.

The assigned column headers were:

symboling, normalized-losses, make, fuel-type, aspiration, num-of-doors, body-style, drive wheels, engine-location, wheel-base, length, width, height, curb-weight, engine-type, num-of-cylinders, engine-size, fuel-system, bore, stroke, compression-ratio, horsepower, peak-rpm, city-mpg, highway-mpg, price

After assigning the column names, the dataset became easier to understand and analyze. Each column represents a specific feature of a car. For example, engine-size, horsepower, city-mpg, and highway-mpg are numerical attributes, while make, fuel-type, body-style, and engine-type are categorical attributes.

1.2. Dataset Shape and First 10 Rows

After loading the dataset, the shape of the dataset was checked to understand the number of observations and variables. The dataset contains 26 column features and 205 rows with column header. This means the dataset includes information about 204 cars and 26 features.

The first 10 rows ([Figure 1](#)) of the dataset were displayed to inspect the structure of the data. This step helped to understand how the values are stored in each column and whether the assigned column names matched the actual data.

1.3. Descriptive Statistics

Descriptive statistics ([Figure 2](#)) were calculated to summarize the numerical columns in the dataset. This step provides an overall understanding of the central tendency, variation, and range of the numerical features.

The main descriptive statistics used were Mean, Median, Standard deviation, Minimum, Maximum, and Quartiles. The `describe()` function in pandas was used to calculate these statistics for numerical columns.

	symboling	normalized-losses	make	fuel-type	aspiration	num-of-doors	body-style	drive-wheels	engine-location	wheel-base	...	engine-size	fuel-system	bore	stroke	compression-ratio	horsepower	peak-rpr
0	3	?	alfa-romero	gas	std	two	convertible	rwd	front	88.6	...	130	mpfi	3.47	2.68	9.0	111	500
1	3	?	alfa-romero	gas	std	two	convertible	rwd	front	88.6	...	130	mpfi	3.47	2.68	9.0	111	500
2	1	?	alfa-romero	gas	std	two	hatchback	rwd	front	94.5	...	152	mpfi	2.68	3.47	9.0	154	500
3	2	164	audi	gas	std	four	sedan	fwd	front	99.8	...	109	mpfi	3.19	3.40	10.0	102	550
4	2	164	audi	gas	std	four	sedan	4wd	front	99.4	...	136	mpfi	3.19	3.40	8.0	115	550
5	2	?	audi	gas	std	two	sedan	fwd	front	99.8	...	136	mpfi	3.19	3.40	8.5	110	550
6	1	158	audi	gas	std	four	sedan	fwd	front	105.8	...	136	mpfi	3.19	3.40	8.5	110	550
7	1	?	audi	gas	std	four	wagon	fwd	front	105.8	...	136	mpfi	3.19	3.40	8.5	110	550
8	1	158	audi	gas	turbo	four	sedan	fwd	front	105.8	...	131	mpfi	3.13	3.40	8.3	140	550
9	0	?	audi	gas	turbo	two	hatchback	4wd	front	99.5	...	131	mpfi	3.13	3.40	7.0	160	550

10 rows × 26 columns

Figure 1. First ten rows of the dataset

1.4. Data Types Identification

The data types of all columns were checked using pandas. This step is important because machine learning models require correct data types. The dataset contains both numerical and categorical variables. There are 16 numerical features and 10 categorical columns.

	symboling	wheel-base	length	width	height	curb-weight	engine-size	compression-ratio	city-mpg	highway-mpg
count	205.000000	205.000000	205.000000	205.000000	205.000000	205.000000	205.000000	205.000000	205.000000	205.000000
mean	0.834146	98.756585	174.049268	65.907805	53.724878	2555.565854	126.907317	10.142537	25.219512	30.751220
std	1.245307	6.021776	12.337289	2.145204	2.443522	520.680204	41.642693	3.972040	6.542142	6.886443
min	-2.000000	86.600000	141.100000	60.300000	47.800000	1488.000000	61.000000	7.000000	13.000000	16.000000
25%	0.000000	94.500000	166.300000	64.100000	52.000000	2145.000000	97.000000	8.600000	19.000000	25.000000
50%	1.000000	97.000000	173.200000	65.500000	54.100000	2414.000000	120.000000	9.000000	24.000000	30.000000
75%	2.000000	102.400000	183.100000	66.900000	55.500000	2935.000000	141.000000	9.400000	30.000000	34.000000
max	3.000000	120.900000	208.100000	72.300000	59.800000	4066.000000	326.000000	23.000000	49.000000	54.000000

Figure 2. Descriptive statistics of the numerical features.

Some numerical columns were initially read as object type because they contained missing values represented by "?". After replacing "?" with NaN, these columns were converted into numeric format.

Categorical columns contain text-based values. For example, body-style includes categories such as sedan, hatchback, wagon, hardtop, and convertible. Other categorical features such as make, aspiration, num-of-doors, drive-wheels, engine-type, and fuel-system also contain different unique categories.

1.5. Visualizations: Histograms

A histogram (Figure 3) of the price column was created to understand the distribution of car prices. From the histogram, it can be observed that the car price distribution is not perfectly normal. Most cars are concentrated in the lower to medium price range, while fewer cars have very high prices. This indicates that the dataset may contain some expensive cars that act as high-value observations or possible outliers.

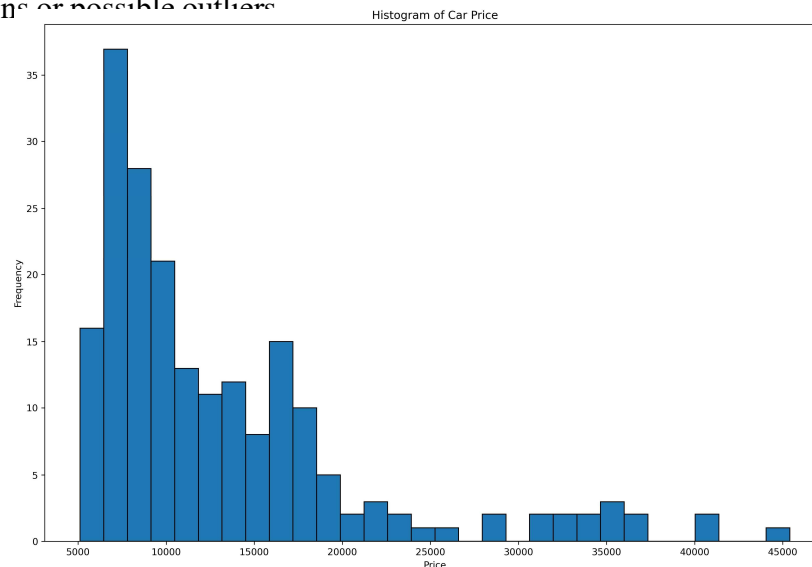


Figure 3. Histogram of the car price distribution.

Most cars have prices between \$5000 and \$20000. Only a small number of cars have prices above \$20000.

A histogram (Figure 4) of the horsepower column was created to understand the distribution of car engine power. The horsepower histogram shows how engine power is distributed across the cars in the dataset. Most cars appear to have low to medium horsepower values, while fewer cars have very high horsepower.

Visualization: Boxplot of Price vs Body Style

A boxplot (Figure 5 (left)) was created to compare car price distributions across different body styles. The body-style column contains categories such as: Sedan, hatchback, wagon, hardtop, convertible. The boxplot shows how prices vary by body style. It helps identify which body styles tend to have higher or lower prices. The median price of hardtop body-style is higher than others and has a wider price range. Some body-style categories contain outliers.

This visualization is useful because body style is a categorical feature, and the boxplot allows comparison of numerical price values across different categories. From a business perspective, this helps identify whether certain body styles are associated with premium pricing.

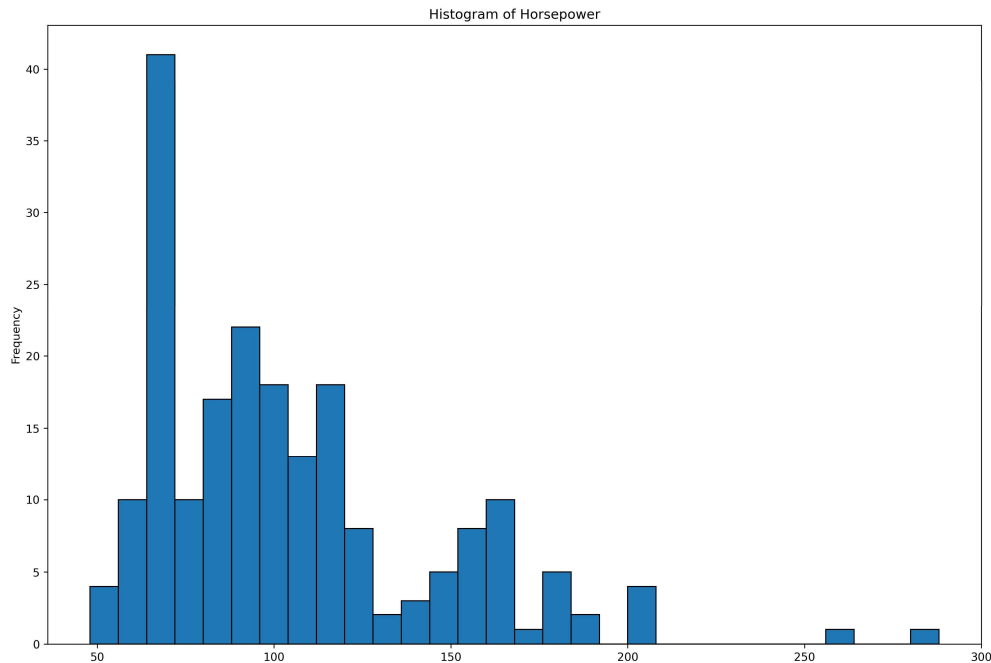


Figure 4. Histogram of the horsepower distribution of the cars.

Visualization : Correlation Heatmap

A correlation heatmap (Figure 5 (right)) was created to examine the relationships between numerical variables. Correlation values range from -1 to +1. +1 means a strong positive relationship, -1 means a strong negative relationship, and 0 means little or no linear relationship. The heatmap helps identify which numerical features are strongly related to price. Positive correlation features are length, wheel-base, engine-size, curb-weight. These features may increase as car price increases. Negative correlations with price include: city-mpg, highway-mpg features. This means that cars with better fuel efficiency may sometimes have lower prices, while larger and more powerful cars may have higher prices.

1.6. Code Explanation:

In the EDA section, different visualizations were created to understand the structure and patterns of the dataset. Histograms were used for price and horsepower to observe their distributions and identify whether most cars fall into low, medium, or high value ranges. A boxplot was created for price against body-style to compare how car prices vary across different body types. A correlation heatmap was also used to examine relationships among numerical features and identify which variables have stronger positive or negative relationships with car price.

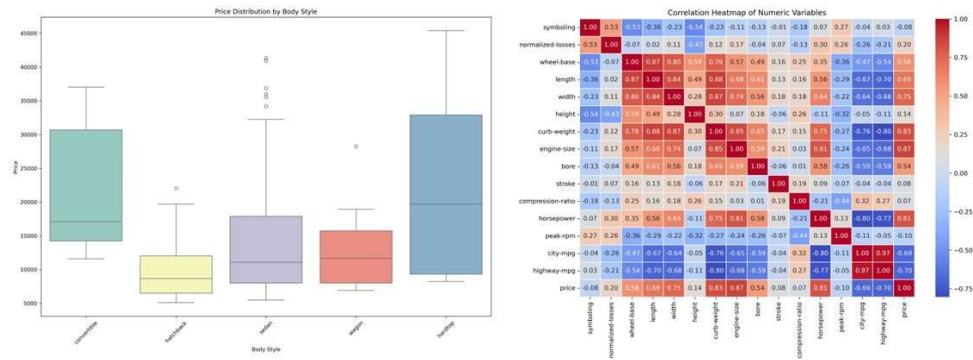


Figure 5. Boxplot of price distribution by body style (left) and heatmap of numerical features (right)

CHAPTER THREE: Data Preprocessing and Feature Engineering

3.1. Handling Missing Values

The dataset contained missing values represented by the symbol "?". Since pandas does not automatically treat "?" as a missing value, these values were first replaced with NaN. After that, missing columns were identified using missing value counts. For numerical columns, missing values were replaced with the mean because it preserves the overall average of the data. For categorical columns, missing values were replaced with the mode, which represents the most frequent category.

3.2. Data Type Conversion

Some numerical columns, such as price and horsepower, were initially stored as object/string data types because of missing symbols like "?". These columns were converted into numeric format using pandas. The price column was converted to float because it is the target variable for regression, while horsepower was converted to float because it is an important numerical predictor for car performance and price.

3.3. Feature Engineering

A new feature called city-L/100km was created from the existing city-mpg column. The formula used was $235 / \text{city-mpg}$. This converted fuel efficiency from miles per gallon into liters per 100 kilometers. This transformation makes fuel consumption easier to interpret in many international contexts, especially where L/100km is commonly used.

3.4. Binning Horsepower

The horsepower column was divided into three categories: Low, Medium, and High. This process is called binning. Binning helps simplify continuous numerical data and makes it easier to compare groups of cars based on engine power. It is useful for understanding how different horsepower levels relate to car price.

3.5. Feature Normalization

Normalization was applied to numerical features to scale their values into a common range, usually between 0 and 1. This is important because different features have different measurement scales. For example, curb-weight, horsepower, and engine-size have very different value ranges. Normalization helps machine learning models treat features more fairly and prevents large-scale variables from dominating smaller-scale variables.

CHAPTER FOUR: Regression Model Development and Evaluation

Part A: Linear Regression

In this part, a Linear Regression model (Galton, 1886) was developed to predict car prices based on selected numerical features. The target variable was price, while the input features included normalized automobile attributes such as engine size, horsepower, curb weight, wheel base, width, height, mileage-related variables, and other technical specifications.

The dataset was split into training and testing sets using an 80/20 split, where 80% of the data was used to train the model and 20% was used to test the model. This approach helps evaluate how well the model performs on unseen data.

The model was evaluated using three performance metrics: R^2 Score, Mean Absolute Error (MAE), and Mean Squared Error (MSE).

For the 80/20 split, the Linear Regression model achieved:

Table 1. Result for Linear Regression model

R^2 Score	0.809
MAE	2757.48
MSE	14,888,090

The R^2 score of 0.809 means that the Linear Regression model explained approximately 80.9% of the variation in car prices. The MAE value shows that, on average, the model's predictions were around 2757.48 units away from the actual car prices.

Part B: Random Forest Regressor

In the second part, a Random Forest Regressor model (Breiman, 2001) was trained to improve the prediction performance. Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to make more accurate and stable predictions.

For the 80/20 split, the Random Forest Regressor achieved:

Table 2. Result for Random Forest Regressor

R^2 Score	0.938
-------------	-------

MAE	1513.51
MSE	4,823,211

The Random Forest model achieved a much higher R^2 score and lower error values than Linear Regression. The R^2 score of 0.938 means that the model explained approximately 93.8% of the variation in car prices. The MAE value shows that the average prediction error was about 1513.51 units, which is much lower than the Linear Regression model.

This indicates that Random Forest performed better because it was able to capture more complex patterns in the data.

Part C: Experimentation with Different Train-Test Splits

To test model stability, the experiment was repeated using a different train-test split: 70/30. In this split, 70% of the data was used for training and 30% was used for testing.

Therefore, comparing multiple splits gives a better understanding of model reliability.

The performance results for both splits are shown below ([Table 3](#)):

Table 3. Comparison table for both splits

Split	Model	R^2 Score	MAE	MSE
80/20	Linear Regression	0.809	2757.48	14,888,090
80/20	Random Forest	0.938	1513.51	4,823,211
70/30	Linear Regression	0.788	2633.70	14,548,460
70/30	Random Forest	0.902	1657.86	6,695,813

Model Comparison

The results show that Random Forest Regressor performed better than Linear Regression in both train-test splits.

For the 80/20 split, Random Forest achieved the highest R^2 score of 0.938 and the lowest MAE and MSE values. This means it produced more accurate predictions than Linear Regression.

For the 70/30 split, Random Forest again performed better, with an R^2 score of 0.902. Although the performance decreased slightly compared to the 80/20 split, it still remained stronger than Linear Regression.

Linear Regression also performed reasonably well, with R^2 scores above 0.78 in both experiments. However, its error values were higher than Random Forest. This suggests that the relationship between car features and price is not purely linear.

Code Explanation:

For the regression task, the target variable was price, and selected numerical features such as engine size, horsepower, curb weight, mileage, and vehicle dimensions were used as input variables. The dataset was divided into training and testing sets using 80/20 and 70/30 splits. A Linear Regression model was first trained as a baseline model, and then a Random Forest Regressor was trained to capture more complex and non-linear relationships. Both models were evaluated using R^2 Score, Mean Absolute Error, and Mean Squared Error to compare their predictive performance.

CHAPTER FIVE: Classification Model

Objective of the Classification Task

In this task, the continuous price variable was converted into a categorical variable to perform a classification problem. Instead of predicting the exact car price, the goal was to classify each car into one of three price groups:

- Low Price
- Medium Price
- High Price

Creating Price Categories

The price column was divided into three equal groups using `pd.qcut()`. The `qcut()` function divides the data based on quantiles. This means the cars were separated into three groups with approximately similar numbers of observations in each group. The new target variable became the name price-category. So, the classification model tried to predict whether a car is in the Low, Medium, or High price category.

Feature Selection

For the classification model, numerical normalized features were used as input variables. The original price column was not used as an input feature because it was used to create the target variable. Including price as an input would cause data leakage and make the model unfairly accurate.

The selected features included vehicle attributes such as: engine size, horsepower, curb weight, wheel base, length, width, height, city-mpg, highway-mpg, city-L/100km, bore, stroke, peak-rpm, compression, ratio, normalized losses. These features were used to predict the price category of each car.

Logistic Regression Model

A Logistic Regression model was used for the classification task. Logistic Regression is a common classification algorithm used to predict categorical outcomes. Although the name contains “regression,” it is mainly used for classification problems.

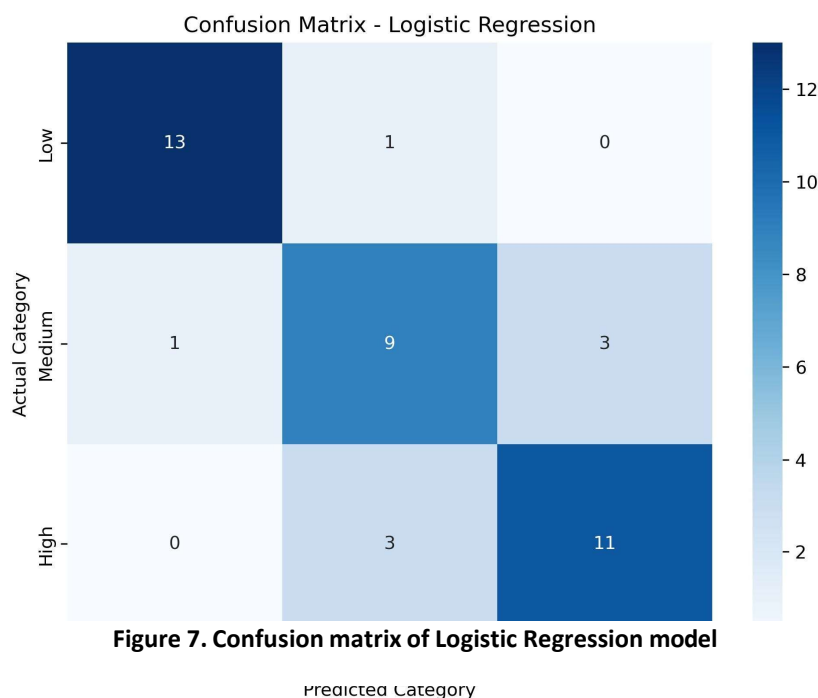
In this project, Logistic Regression was used to classify cars into: low price, medium price, high price. The dataset was split (80/20) into training and testing sets. The training data was used to train the model, and the testing data was used to evaluate how well the model performed on unseen data. The overall F1-score was achieved 80% by Logistic Regression model.

Classification Report:				
	precision	recall	f1-score	support
High	0.79	0.79	0.79	14
Low	0.93	0.93	0.93	14
Medium	0.69	0.69	0.69	13
accuracy			0.80	41
macro avg	0.80	0.80	0.80	41
weighted avg	0.80	0.80	0.80	41

Figure 6. The entire classification report of Logistic Regression Model

Interpretation of the Classification Result

The confusion matrix helps to understand where the model performs well and where it makes mistakes. If most values are on the diagonal, the model is performing well. If many values are outside the diagonal, the model is confusing some classes. For this task, the model was expected to classify cars into three price categories. In general, misclassification may happen between



nearby categories, such as:

- Low and Medium

- Medium and High

This is understandable because cars near the boundary between two price groups may have similar features. For example, a car at the upper end of the Medium category may look similar to a car at the lower end of the High category. Therefore, the model may sometimes confuse these categories.

Importance of the Classification Model

The classification model is useful from a business perspective because it can help group cars into market segments. Instead of predicting an exact price, the model can tell whether a car belongs to a low, medium, or high price segment.

This can help companies:

- identify budget, mid-range, and premium cars
- support pricing strategy
- understand product positioning
- segment customers
- recommend cars based on price level

Customers can also use this type of model to understand whether a car is likely to be in a low, medium, or high price range based on its features.

Code Explanation:

For the classification task, the continuous price column was converted into three categories: Low, Medium, and High using `pd.qcut()`. This function divides the price values into three approximately equal groups based on quantiles. After creating the new price-category target variable, a Logistic Regression model was trained to classify cars into these price groups. The model was evaluated using accuracy and a confusion matrix, which showed how many cars were correctly or incorrectly classified in each price category.

CHAPTER SIX: Model Evaluation and Visualization

Visual Evaluation of Actual vs Predicted Prices

Actual vs Predicted price plots (Figure 9) were also used to visually compare model performance. In these plots, the x-axis represents the actual car prices, and the y-axis represents the predicted car prices. The diagonal reference line represents perfect prediction.

For a good model, the points should be close to the diagonal line. The Random Forest plots showed that the predicted values were closer to the actual values compared to Linear Regression. This supports the numerical evaluation results and confirms that Random Forest performed

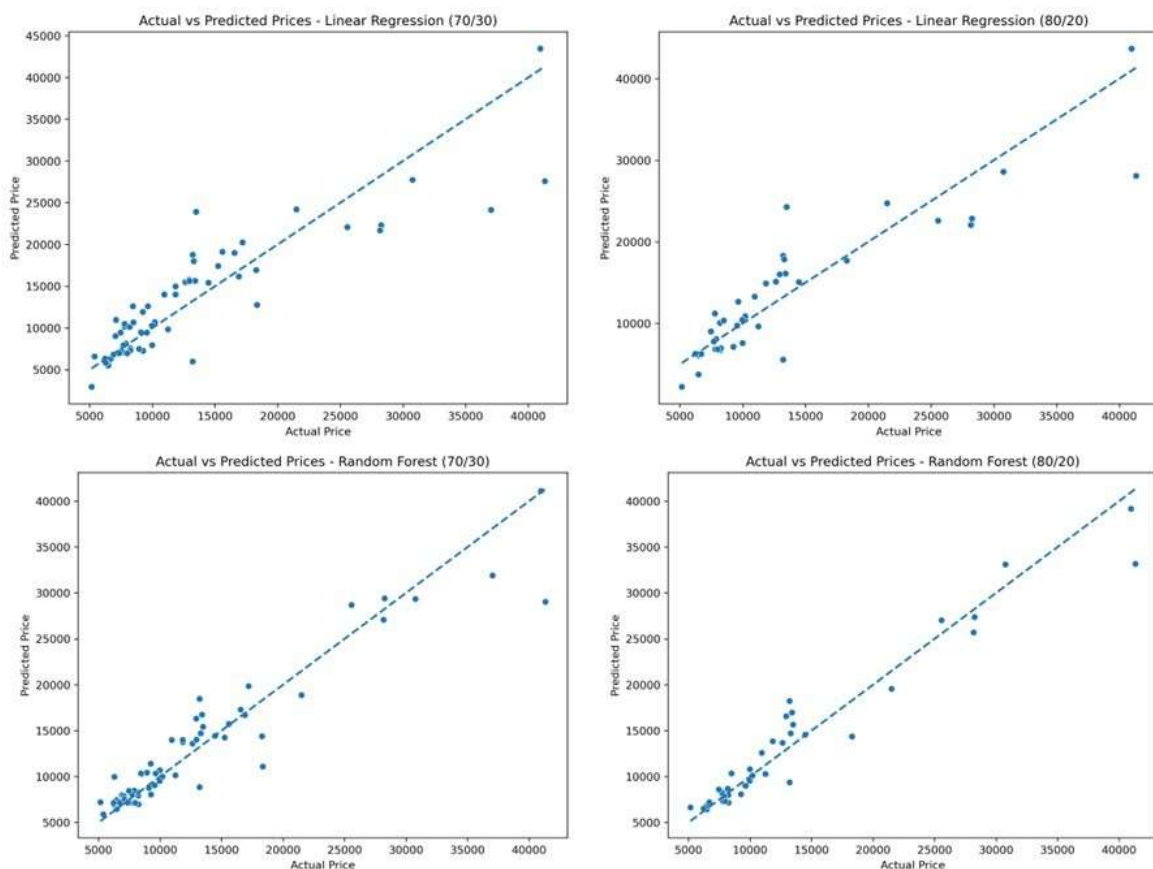


Figure 8. Actual vs predicted values for Linear Regression and Random Forest models in both splits

better.

CHAPTER SEVEN: Business Insights & Recommendations

What makes a car expensive?

Based on the model results, the most important factors that make a car expensive are engine size, curb weight, highway mileage, horsepower, and car width. Among these, engine size and curb weight were the strongest predictors. This means that cars with larger engines and heavier body structures usually have higher prices. These features are often related to better performance, stronger build quality, and more premium vehicle design.

Which features should companies focus on?

Car companies should focus on features that strongly influence price, especially engine size, curb weight, horsepower, and fuel efficiency. Engine size and horsepower are important for performance, while curb weight and width are related to vehicle size, comfort, and structure. Fuel efficiency is also important because customers often consider long-term fuel costs before buying a car. Therefore, companies can use these features to design different car segments such as economy cars, mid-range cars, and premium cars.

How can customers use this model?

Customers can use this model to estimate whether a car price is reasonable based on its features. For example, by looking at features such as engine size, horsepower, fuel efficiency, and car weight, the model can predict the expected price range of a car. This can help customers compare different cars, identify overpriced vehicles, and make better purchasing decisions. The classification model can also help customers understand whether a car belongs to a low, medium, or high price category.

CHAPTER EIGHT: Conclusion

This project analyzed an automobile dataset to predict car prices using machine learning models. The target variable was price, and important numerical features such as engine size, curb weight, horsepower, fuel efficiency, and car dimensions were used for prediction.

Two regression models were compared: Linear Regression and Random Forest Regressor. The results showed that Random Forest performed better in both 80/20 and 70/30 train-test splits. The best result was achieved with the 80/20 split, where Random Forest obtained an R^2 score of 0.938, MAE of 1513.51, and MSE of 4,823,211. The most important features were engine size, curb weight, highway MPG, horsepower, and width. This shows that cars with larger engines, heavier weight, and stronger performance-related features tend to be more expensive.

However, the project has some limitations. The dataset was relatively small, and mainly numerical features were used. Important categorical features such as make, fuel type, body style, drive wheels, engine type, and fuel system were not fully included in the final regression model. Missing values were also handled using simple mean and mode replacement.

In the future, the model could be improved by including categorical features through one-hot encoding, applying cross-validation, tuning hyperparameters, detecting outliers, and testing advanced models such as XGBoost, LightGBM, or Gradient Boosting. For classification, car prices were divided into Low, Medium, and High categories, which can help companies and customers understand different car market segments.

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